

Relations

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DM/DS

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Binary Relations

Let P and Q be two non-empty sets. A binary relation R is defined to be a subset of $P \times Q$ from a set P to Q . If $(a, b) \in R$ and $R \subseteq P \times Q$, then a is related to b by R . i.e. $a R b$.

Example 1: Let $A = \{a, b, c\}$, $B = \{r, s, t\}$, then $R = \{(a, r), (b, r), (b, t), (c, s)\}$ is a relation from A to B .

Example 2: If A has m elements and B has n elements, how many relations are there from A to B .

Solution: There are $m \times n$ elements, hence there are $2^{m \times n}$ relations from A to B .

Example 3: If $A = \{1, 2\}$. Determine all relations from A to A .

Solution: There are $2^2 = 4$ elements, i.e. $\{(1, 1), (1, 2), (2, 1), (2, 2)\}$ and there are $2^4 = 16$ relations from A to A .
 $\emptyset, \{(1, 1)\}, \{(1, 2)\}, \{(2, 1)\}, \{(2, 2)\}, \{(1, 2), (2, 1)\}, \{(1, 2), (1, 1)\}, \{(1, 2), (2, 2)\}, \{(2, 1), (1, 1)\}, \{(2, 1), (2, 2)\}, \{(1, 1), (2, 2)\}, \{(1, 2), (2, 1), (1, 1)\}, \{(1, 2), (1, 1), (2, 2)\}, \{(2, 1), (1, 1), (2, 2)\}, \{(1, 2), (2, 1), (2, 2)\}, \{(1, 1), (2, 1), (1, 2), (2, 2)\}.$

Domain of relation: The domain of a relation R is the set of elements in P which are related to some elements in Q or it is the set of all first entries of the ordered pairs in R . It is denoted by $\text{DOM}(R)$.

Range of relation: The range of relation R is the set of elements in Q which are related to some elements in P or it is the set of all second entries of the ordered pairs in R . It is denoted by $\text{RAN}(R)$.

Example 4: Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$ and $R = \{(1, a), (1, b), (1, c), (2, b), (2, c), (2, d)\}$.

Then $\text{DOM}(R) = \{1, 2\}$ and $\text{RAN}(R) = \{a, b, c, d\}$.

Complement of a relation: Consider a relation R from a set A to B . The complement of relation R denoted by $\bar{R}$ is a relation from A to B such that $\bar{R} = \{(a, b) : (a, b) \notin R\}$

Example 5: Let $X = \{1, 2, 3\}$, $Y = \{8, 9\}$ and

$R = \{(1, 8), (2, 8), (1, 9), (3, 9)\}$. Find Complement of R .

Sol: First $X \times Y = \{(1, 8), (1, 9), (2, 8), (2, 9), (3, 8), (3, 9)\}$

$$\text{then } \bar{R} = \{(3, 8), (2, 9)\}$$

Example 6: Let $A = \{7, 8, 9\}$, $B = \{k, l, m, n\}$ and

$R = \{(7, k), (8, k), (8, l), (8, m), (9, m), (9, n)\}$. Find $\bar{R}$.

Sol: $A \times B = \{(7, k), (7, l), (7, m), (7, n), (8, k), (8, l), (8, m), (8, n), (9, k), (9, l), (9, m), (9, n)\}$.

$$\therefore \bar{R} = \{(7, l), (7, m), (7, n), (8, n), (9, k), (9, l)\}$$

Inverse of a relation: Consider a relation R from a set A to B . The inverse of a relation R , denoted

by $\bar{R}$, is a relation from B to A such that

$b \bar{R} a \iff a R b$. i.e.

$$\bar{R} = \{(b, a) : (a, b) \in R\}.$$

Example 7: $A = \{2, 3, 4, 5\}$, $B = \{a, b, c\}$ and

$R = \{(2, a), (2, c), (3, b), (3, c), (4, a), (5, a), (5, b)\}$

$$\bar{R} = \{(a, 2), (c, 2), (b, 3), (c, 3), (a, 4), (a, 5), (b, 5)\}$$

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Representation of relations: Relations can be represented in many ways.

1. Relation as a matrix:

Let $P = \{a_1, a_2, \dots, a_m\}$ and $Q = \{b_1, b_2, \dots, b_n\}$ are finite sets, containing m and n number of elements respectively. R is a relation from P to Q . The relation R can be represented by $m \times n$ matrix $M = M_{ij}$, defined as

$$M_{ij} = \begin{cases} 0 & \text{if } (a_i, b_j) \notin R \\ 1 & \text{if } (a_i, b_j) \in R \end{cases}$$

Ex: Let $P = \{1, 2, 3, 4\}$, $Q = \{a, b, c, d\}$

and $R = \{(1, a), (1, b), (1, c), (2, b), (2, c), (2, d), (3, a), (3, d), (4, b), (4, c)\}$

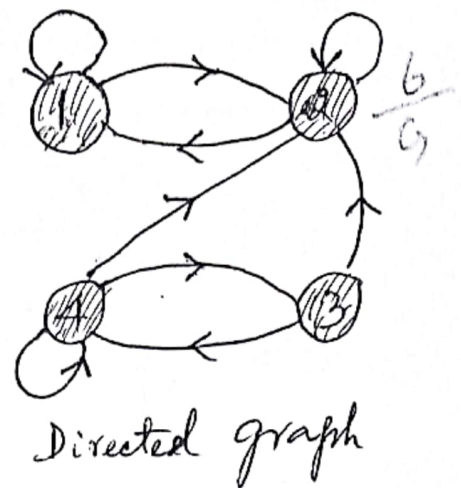
$$M_R = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

2. Relation as a directed graph:

If P is a finite set and R is a relation on P , R can be represented as a directed graph.

Ex. Let $P = \{1, 2, 3, 4\}$

$$R = \{(1,1), (1,2), (2,1), (2,2), (3,2), (3,4), (4,3), (4,4), (4,2)\}$$



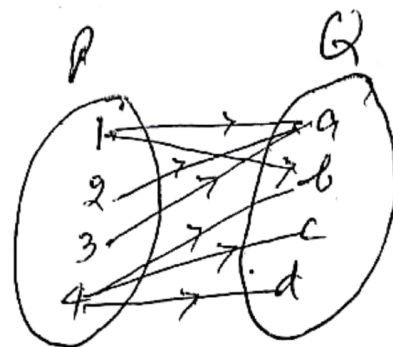
3. Relation as a Arrow diagram:

If P and Q are finite set and R is a relation from P to Q . Relation R can be represented as an arrow diagram.

e.g: Let $P = \{1, 2, 3, 4\}$

$$Q = \{a, b, c, d\}$$

and $R = \{(1,a), (2,a), (3,a), (1,b), (4,b), (4,c), (4,d)\}$



Properties of relations:

Property of the relation tells the nature or type of the relation. There are many properties of relations i.e Reflexive, Symmetric, Transitive, anti-symmetric etc.

Reflexive relation: Consider a binary relation R on a set A . Relation R is called a reflexive relation if, for every a in set A , $(a, a) \in R$.

Example: $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (1, 3), (1, 2), (2, 3), (3, 3)\}$

then R is reflexive relation.

2. Symmetric relation: Consider a binary relation R on a set A . R is called symmetric if $(a, b) \in R$ or $\longrightarrow (b, a) \in R$.

Ex: $A = \{1, 2, 3\}$, $R = \{(1, 1), (1, 2), (2, 1), (1, 3), (3, 1), (3, 2), (2, 2), (2, 3)\}$.

then R is Symmetric relation.

3. Transitive relation: Consider the binary operation R on a set A . R is called transitive relation if whenever $(a, b) \in R$ and $(b, c) \in R$ or $\longrightarrow (a, c) \in R$.

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Ex: Let $A = \{1, 2, 3\}$ & $R = \{(1, 2), (2, 2), (2, 1), (1, 1)\}$
then R is transitive relation.

Ex: Let $A = \{3, 4, 5\}$ $R = \{(3, 4), (4, 3), (5, 4), (5, 3), (3, 3)\}$
Relation is not transitive since $(4, 3) \in R$ and $(3, 4) \in R$
but $(4, 4) \notin R$.

4. Anti-Symmetric relation

Consider a binary relation R on Set A . Relation

R is called anti-symmetric relation if

$(a, b) \in R \Rightarrow (b, a) \notin R$ unless $a = b$.

or if (a, b) & (b, a) belongs to R , then $a = b$.

Ex: Let $A = \{4, 5, 6\}$, $R = \{(4, 4), (4, 5), (5, 6), (4, 6)\}$
then R is Anti Symmetric Relation

Ex: Let $A = \{4, 5, 6\}$, $R = \{(4, 4), (4, 5), (5, 6), (6, 5), (4, 6)\}$
then R is not anti-symmetric since
 $(5, 6) \in R$ & $(6, 5) \in R$ but $6 \neq 5$.

Example: $A = \{1, 2, 3, 4\}$, $R = \{(1,1), (2,2), (1,2), (2,1), (4,1), (2,4), (3,3), (1,3), (1,4)\}$ $\frac{9}{6}$

Then R is Symmetric but not Reflexive and ^{not} Transitive

Example: $A = \{4, 5, 6\}$, $R = \{(4,4), (4,5), (4,6), (5,4), (5,5), (6,6),$

R is Reflexive, Not Symmetric, Transitive, Not anti Symmetric

Closure properties of relation

Consider a relation R on Set A . Suppose the relation R does not possess the desired property. we will add as few new pairs as possible to relation R to get a relation that does have the desired property.

The smallest relation S on A that contains R and the desired property is called the Closure of relation with the desired property.

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Equivalence Relation: A relation R on a set A is called an equivalence relation if it satisfies following three properties:

- (i) Relation is reflexive i.e. $aRa, \forall a \in A$.
- (ii) Relation is Symmetric i.e. $aRb \Rightarrow bRa$.
- (iii) Relation is transitive i.e. $aRb \wedge bRc \Rightarrow aRc$.

Ex: Let $A = \{1, 2, 3, 4\}$.

$$R = \{(1,1), (1,3), (2,2), (2,4), (3,1), (3,3), (4,2), (4,4)\}$$

then R is reflexive: $\forall a \in A \Rightarrow aRa \text{ or } (a,a) \in R$.

R is Symmetric: $(a,b) \in R \Rightarrow (b,a) \in R$.

R is transitive: $(a,b) \wedge (b,c) \in R \Rightarrow (a,c) \in R$.

$\therefore R$ is an equivalence relation.

Ex: Let $A = \{0, 1, 2, 3\}$

$$R = \{(0,0), (1,1), (1,2), (2,3), (2,2), (3,2), (3,3), (1,0), (3,1), (2,1), (1,3), (0,2), (0,3), (2,0)\}$$

then R is Eq. Relation.

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Partial Order Relation:

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A relation R on a set A is called a Partial Order relation if it satisfies the following three properties:

1. Reflexive i.e. $aRa, \forall a \in A$
2. Anti Symmetric aRb then $b \nsubseteq a$ or $aRb \& bRa \Rightarrow a=b$.
3. Transitive i.e. $aRb \& bRc \Rightarrow aRc$.

Ex: Given an example show that the relation $(x, y) \in R$ if $x \geq y$ defined on the set A of +ve integers is a partial order relation.

Sol: Let $A = \{1, 2, 3, 4\}$ and $x \geq y$, then

$$R = \{(1,1), (2,2), (3,3), (4,4), (2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$$

1. Reflexive: $\forall a \in A, (a,a) \in R$ i.e. $(1,1), (2,2), (3,3), (4,4) \in R$.
 2. Anti Symmetric: $(a,b) \in R$ then $(b,a) \notin R$ or $(a,b) \in R \& (b,a) \in R \Rightarrow a=b$.
 3. Transitive: $(a,b) \in R \& (b,c) \in R \Rightarrow (a,c) \in R$
i.e. $(4,2) \in R \& (2,1) \in R \Rightarrow (4,1) \in R$.
- $\therefore$ P. O. Relation

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Ex: $(x, y) \in R$ if $x \leq y$ defined on the integer
is a p.o. relation. $\frac{14}{6}$

Ex: $A = \{3, 4, 5\}$, $R = \{(3, 4), (4, 3), (5, 4), (5, 3), (3, 3)\}$

- (i) then R is not reflexive: since $(4, 4), (5, 5) \notin R$.
(ii) then R is not symmetric: since $(5, 4) \in R$ but $(4, 5) \notin R$.
(iii) then R is not antisymmetric: since $(3, 4) \& (4, 3) \in R$ but $3 \neq 4$.
(iv) then R is not Transitive: $(a, b) \in R \& (b, c) \in R \Rightarrow (a, c) \notin R$
i.e. $(5, 4) \& (4, 3) \in R \Rightarrow (5, 3) \notin R$.